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Anesthetic Management of Awake Craniotomy for Resection of the Language and Motor Cortex Vascular Malformations

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BACKGROUND: **Although the safety and feasibility of awake craniotomy are well established for epilepsy and brain tumor surgery, its application for resection of vascular lesions, including arteriovenous malformations (AVMs) and cavernomas, is still limited. Apart from the usual challenges of awake craniotomy, vascular lesions pose several additional problems. Our goal is to determine the safety and practicality of awake craniotomy in patients with cerebral vascular malformations located near the eloquent areas, using a refined anesthetic protocol.**

METHODS: **A retrospective case series was performed on 7 patients who underwent awake craniotomy for resection of AVMs or cavernomas located in the eloquent language and motor areas. Our protocol consisted of achieving deep sedation, without a definitive airway, using a combination of propofol, dexmedetomidine, and remifentanil/fentanyl during scalp block placement and surgical exposure, then transitioning to a wakeful state during the resection.**

RESULTS: **Six patients had intracranial AVMs, and 1 patient had a cavernoma. Six patients had complete resection; however, 1 patient underwent repeat awake craniotomy for residual AVM nidus. The patients tolerated the resection under continuous awake neurologic and neurophysiologic testing without significant perioperative complications or the need to convert to general anesthesia with a definitive airway.**

CONCLUSIONS: **Awake craniotomy for excision of intracranial vascular malformations located near the eloquent areas, in carefully selected patients, can facilitate resection by allowing close neuromonitoring and direct functional assessment. A balanced combination of sedative and analgesic medications can provide both adequate sedation and rapid wakeup, facilitating the necessary patient interaction and tolerance of the procedure.**

INTRODUCTION

Cerebral arteriovenous malformations (AVMs) comprise a complex tangle of abnormal arteries and veins, which can be a significant source of cerebral hemorrhage. Cerebral AVMs situated in eloquent areas (e.g., the language, motor, and visual cortices) pose substantial diagnostic and therapeutic challenges. Given the potential for neurologic deterioration with surgical intervention,[1,2] asymptomatic patients are often managed conservatively or with minimally invasive procedures.[3,4] However, surgical intervention may be warranted in the setting of seizures or other symptoms caused by local mass effect or AVM rupture.[4]

The management of cerebral AVMs located in eloquent areas has evolved over the years.

Key words
- Arteriovenous malformations
- Brain vascular malformations
- Craniotomy
- Dexmedetomidine
- Propofol
- Remifentanil

Abbreviations and Acronyms
AAA: Asleep-awake-asleep
AVM: Arteriovenous malformation
BP: Blood pressure
CT: Computed tomography
DCS: Direct cortical stimulation
DSA: Digital subtraction angiography
fMRI: Functional magnetic resonance imaging
GA: General anesthesia
MAC: Monitored anesthetic care
MRI: Magnetic resonance imaging
OSA: Obstructive sleep apnea
POD: Postoperative day
SMG: Spetzler-Martin grade

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Citation: World Neurosurg. (2020) 143:e136-e148. https://doi.org/10.1016/j.wneu.2020.07.050

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

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Among prevailing treatment modalities, endovascular embolization is most commonly used as adjunctive treatment to surgery or radiosurgery; it is infrequently used as primary therapy.[4,5] Stereotactic radiosurgery is another option, with a cure rate of approximately 64%. Complete occlusion with radiosurgery is often delayed, with significant risk of bleeding during the latency period.[4,6,7] However, microsurgical excision can achieve immediate angiographic cure in approximately 98% of cases,[8,9] minimizing the risk of future hemorrhage or need for long-term follow-up. The larger AVMs in eloquent locations carry significant surgical risks not fully elucidated in the neurovascular literature.[10,11]Association with the eloquent cortex itself carries an unfavorable prognosis,[1,10-13] including prolonged hospitalization secondary to neurologic deficits incurred after surgery.[1] Therefore, safe resection of AVMs, without causing any additional neurologic deficits, is paramount. Awake resection of the lesion under close neurologic monitoring is promising in achieving this goal.

Awake craniotomy is a well-established technique for epilepsy and brain tumor surgery near the eloquent areas.[14-16] However, the evidence for its application in resection of eloquent vascular lesions is mostly limited to small case series.[17-22] The previous case series have not described the specifics of anesthetic management or drug regimen used. This technique requires an expert team of neuroanesthesiologists because the quality of anesthetic management and patient cooperation play a crucial role.

Through a retrospective case series, we describe the anesthetic management of 7 patients who underwent successful awake craniotomies for excision of intracranial vascular malformations located near the eloquent language and motor areas, along with a review of the literature. A refined anesthetic protocol using a combination of propofol, dexmedetomidine, and remifentanil/fentanyl was used to maintain deep sedation without the need for intubation during surgical exposure, allowing for rapid awakening before surgical resection.

METHODS

Data Collection

A retrospective case series was performed after obtaining institutional review board exemption at Loma Linda University Medical Center and Children's Hospital in compliance with Health Insurance Portability and Accountability Act regulations. Included in the case series were patients who underwent awake surgery for AVMs or cavernoma located within or adjacent to the language and/or motor cortex based on magnetic resonance imaging (MRI) from February 2018 to May 2020. We obtained the following data points: clinical indications, presenting symptoms, patient demographics, relevant medical history, preoperative laboratory results, preoperative and postoperative imaging (computed tomography [CT], MRI, magnetic resonance angiography, digital

Table 1. Patient Demographics

Patient	Age (years)	Sex	Past Medical History	Presenting Symptoms	SMG	Location and Nidus Size (cm)	Indication for Awake Surgery
1	46	M	Prior ICH due to AVM rupture causing left-sided weakness, HTN, HLD	Headache, nausea, intermittent blurry vision, forgetfulness, increasing left-sided weakness, seizures	Ruptured AVM grade 2	Right frontoparietal (2.2 × 2.2)	Nearby eloquent motor function
2	50	M	Depression, former smoker	Occipital headache with lower neck stiffness, photophobia, nausea, spasm, difficulty with reading comprehension, aphasia	AVM grade 3	Left temporal cortex (4.0 × 3.0)	Nearby eloquent language and motor function
3	17	M	Seizures— partial symptomatic epilepsy with complex partial seizures	Unprovoked seizure. Left hand tremors, left eye pain and blurriness	AVM grade 2	Right precentral gyrus (2.9 × 2.7)	Nearby eloquent motor function
4	14	F	Previous ICH caused by AVM rupture	Constant left-sided headache, acutely worsened with nausea and emesis, numbness in right hand, with slurring of speech	Ruptured AVM grade 2	Left temporoparietal cortex (2.5 × 0.6)	Nearby language area
5	33	F	Headache, migraines, cholelithiasis, obesity	Severe headache and migraine	Cavernoma	Left temporoparietal cortex (1.7 × 1.7)	Nearby eloquent language and motor function
6	22	M	Asthma, seizures, obesity	Seizures and focal weakness in left leg	AVM grade 2	Right paramedian posterior frontal lobe (2.5 × 2.1)	Nearby eloquent motor function
7	50	M	HTN, gastroesophageal reflux disease, smoking, alcohol and drug abuse	Right-hand intermittent weakness and numbness, speech difficulties, seizures	AVM grade 2	Left frontoparietal cortex (2.5 × 2.3 × 1.8)	Nearby eloquent motor function

M, male; ICH, intracranial hemorrhage; AVM, arteriovenous malformation; HTN, hypertension; HLD, hyperlipidemia; F, female; SMG, Spetzler-Martin grade.

subtraction angiography [DSA]), intraoperative anesthetic techniques and neuromonitoring/mapping details, intraoperative hemodynamic parameters, interventional embolization or coiling, complications, and clinical outcomes (Tables 1–3).

All patients had CT and/or CT angiography and standard MRI preoperatively. In this context, standard MRI includes sagittal T1 with and without contrast, T2 axial and coronal images, T2 fluid-attenuated inversion recovery, susceptibility-weighted imaging, and diffusion-weighted images. When indicated, magnetic resonance angiography and functional MRI (fMRI) were obtained to localize the lesion in relation to the eloquent areas. All patients with AVM also underwent DSA and, if deemed suitable, preoperative embolization under general anesthesia (GA) before surgery.

Anesthetic Management

Intraoperatively, all patients were monitored with continuous electrocardiography, invasive arterial line, end-tidal CO_2, pulse oximetry, temperature monitor, and indwelling urinary catheter. Supplemental oxygen was provided by face mask. An asleep-awake-asleep (AAA) technique was used in all patients. Deep sedation was achieved using continuous infusions of propofol (30–150 μg/kg/minute), dexmedetomidine (0.2–1.4 μg/kg/hour), and/or remifentanil (0.03–0.08 μg/kg/minute). For the pediatric patient, boluses of fentanyl were added instead of remifentanil and titrated to effect. The neurosurgeon performed the scalp block, using a mixture of lidocaine 0.5% with epinephrine 1:200,000 and 0.5% bupivacaine. Local anesthetic was also infiltrated along the incision and at the pin sites. A nasopharyngeal airway was lubricated with lidocaine ointment and inserted into 1 or both nostrils as needed to maintain airway patency. Intravenous acetaminophen, famotidine, dexamethasone, and ondansetron were administered prophylactically to all patients. After achieving deep sedation, patients were carefully positioned, with the face toward the anesthesiology team, and drapes secured to create appropriate access. Care was taken to achieve optimal operative

Table 2. Intraoperative and Postoperative Results

Patient Number	Body Mass Index (kg/m^2)	Obstructive Sleep Apnea Risk	Estimated Blood Loss (mL)	Intraoperative Complications	Discharge Day (Postoperative Day)	Immediate Postoperative Complications	Long-Term Follow-Up Results	Angiography and MRI Results
1	29.3	High	400	None. Bilateral nasal airway to maintain airway patency	3	None	Neurologically stable at 18 months postoperatively	Complete excision
2	28.3	High	900	Required intraoperative transfusion, bilateral nasal airway for desaturation	4	None	Stable at 13 months. Improvement in reading comprehension at 1 year follow-up	Complete excision
3	22.6	Low	1350	Blood loss needing postoperative transfusion	4	Mild facial weakness, which recovered in 3 days	Seizure at 3 months because of residual arteriovenous malformation required second awake surgery	Complete excision confirmed with angiography after the second surgery
							No new seizures since the second surgery	
							Asymptomatic at 6 months since the second surgery	
4	20.5	Low	50	Anxiety and nausea toward the end requiring sedation for the final part of excision	4	None	Stable at 12 months	Complete excision
5	39	High	50	Desaturation responsive to nasal airway insertion	1	Mild headache	Stable at 13 months	MRI showing complete cavernoma resection
6	33.4	High	400	Bilateral nasal airway to maintain airway patency. 1 episode of seizure terminated quickly. Left-sided weakness during the final part of resection	6	Left-sided weakness. Recovered completely in 7 days	Stable at 4 months	Complete excision
7	28.5	High	200	Bilateral nasal airway to maintain airway patency during asleep portion	4	None	Stable at 10 days	Complete excision

MRI, magnetic resonance imaging.

Table 3. Summary of Intraoperative Hemodynamics, Drug Infusion Doses, and Timing

	Preinduction		Asleep					Awake				Asleep					
Patient Number	HR Range (bpm)	SBP/DBP Range (mm Hg)	DEX (μg/kg/hour)	PROP (μg/kg/minute)	REMI (μg/kg/minute)	HR Range (bpm)	SBP/DBP Range (mm Hg)	Time to Awakening (minutes)	Duration (hours)	HR Range (bpm)	SBP/DBP Range (mm Hg)	DEX (μg/kg/hour)	PROP (μg/kg/minute)	REMI (μg/kg/minute)	HR Range (bpm)	SBP/DBP Range (mm Hg)	Total Operating Room Time (hours)
1		121—152					100—136			89—106	114—127					93—115	
	73—90	42—83	0.5—1.1	40—75	0.05—0.07	52—81	53—66	6	3.25		55—64	0.3–0.6	20	0.07	58—79	45—62	7
		118—129				51—69	77—96				82—109					86—91	
2	60—72	60—82	0.2—1.0	40—50	0.03—0.08		32—69	15	7.75	60—68	58—66	0.2—0.7	20—30	0.01—0.04	50—63	56—60	14
		100—115				87—97	79—98				102—116					88—111	
3	100—120	42—60	0.3—1.0	20—75	0.03—0.05		37—51	15	6.3	95—104	49—59	0.2—0.4	10—50	0.01—0.05	92—100	49—61	9
		130—152				52—67	90—116				110—124					80—98	
4*	70—96	78—90	0.5—0.7	50—200	——		40—65	13	5	70—84	54—67	0.5	10—20	——	56—78	49—62	7
		128—145				89—98	93—106				102—134					109—124	
5	102—120	85—96	0.5—1.0	50—100	0.03—0.08		55—72	8	2	88—112	65—87	0.1	30—50	0.03	86—99	70—80	4
		120—133				92—102	81—115				95—118					91—107	
6	101—124	57—61	0.2—0.7	30—80	0.03—0.06		46—65	6	4.75	85—100	41—58	0.2—0.4	40—60	0.03—0.04	84—93	39—51	7.3
		141—150					95—139				95—124					94—109	
7	74—77	86—96	0.2—1.4	30—70	0.03—0.08	47—85	57—70	10	3.5	63—92	51—67	0.3	30—50	0.03—0.04	58—78	49—59	6.5

HR, heart rate; SBP, systolic blood pressure; DBP, diastolic blood pressure; DEX, dexmedetomidine; PROP, propofol; REMI, remifentanil.
*Patient 4 received intermittent fentanyl boluses (total 200 μg).

positioning for the surgeon and to maintain adequate jugular venous drainage and airway patency. Image-guided navigation was used in all cases. Deep sedation was maintained during the craniotomy and exposure. Mannitol 0.5–1 mg/kg was administered to achieve brain relaxation as needed. Once the lesion was exposed, sedation was discontinued, and patients were awakened. In some patients, low-dose remifentanil (0.01–0.03 μg/kg/minute) and/or dexmedetomidine (0.1–0.2 μg/kg/hour) infusions were restarted during part of the awake phase for patient comfort, maintaining adequate wakefulness to participate in neurologic examination. Patients were re-sedated for closure after the excision.

Cortical Stimulation Mapping

Neuromonitoring consisted of somatosensory-evoked potentials, electroencephalography, and direct cortical stimulation (DCS) for mapping language and motor functions. DCS was performed with the Ojemann Cortical Stimulator (Radionics, Burlington, Massachusetts, USA). Starting at 2 mA, the cortical regions surrounding the lesion were stimulated at 0.5-cm to 1-cm intervals up to 10 mA, along with continuous intracranial electrocorticography. Sterile numbered labels were used to identify the stimulation sites. Motor areas were identified by stimulation-triggered movement; language mapping was performed by asking patients to count numbers, name objects from pictures, and read words from flash cards or a screen; speech areas were identified by stimulation-induced speech arrest, errors in counting, naming, or reading words. If the perimalformation zone showed negative language and motor response, we did not always look for positive responses. On conclusion of cortical mapping, the lesion was resected while keeping the patient awake. Temporary occlusion of the large feeding arteries was performed as needed, while monitoring for any neurologic deterioration. Subcortical stimulation was carried out around the nidus when the anatomy implied closeness to eloquent white matter tracts. Neurologic examination was repeated throughout the surgery until the lesion was completely excised.

Postoperatively, all patients had CT head imaging; all patients with AVM also had postoperative DSA. Patients were subsequently followed in the neurosurgery clinic to monitor recovery and evaluate for long-term outcomes.

RESULTS

Six patients had intracranial AVMs 1 patient had a cavernoma, with all lesions located proximal to the eloquent language and motor areas. The patients ranged in age from 14 to 50 years (Table 1). Two patients had previous intracranial hemorrhage from AVM rupture. Three patients had lesions near eloquent language areas, and 4 patients had lesions near the motor area. Five patients had Spetzler-Martin grade (SMG) 2 AVMs, and 1 patient had SMG 3 AVM. All patients with AVM underwent preoperative cerebral angiography under GA. Four of these patients underwent preoperative embolization of feeder arteries to the AVMs; however, embolization was not possible in 2 patients.

During awake craniotomy, patients were initially sedated with propofol, dexmedetomidine, and remifentanil infusions, and 1 patient received fentanyl boluses instead of remifentanil (Table 3). All patients had smooth awakening after cessation of the infusions and were cooperative for neurologic examination within an average time of 10 minutes (range, 6–15 minutes). All patients smoothly transitioned from deep sedation to wakefulness without coughing or laryngospasm. Five patients had risk factors for obstructive sleep apnea (OSA) and experienced airway obstruction during deep sedation, which subsided after inserting bilateral nasopharyngeal airways. Intraoperative arterial blood gas analysis confirmed adequate oxygenation and ventilation. Patient 6 had an intraoperative seizure, which responded to cold lactated Ringer irrigation. All patients remained cooperative throughout the resection, except for the pediatric patient, who had some intraoperative anxiety and nausea during the latter part of the excision and required sedation through the end of the case. Blood loss ranged from 50 to 1350 mL. Two patients had significant bleeding requiring blood transfusion. Patients remained hemodynamically stable throughout surgery, defined as maintaining a target blood pressure (BP) within 20% of baseline values (see Table 3). Nicardipine and phenylephrine infusions were available if needed. Five patients (patients 1, 3, 5, 6, and 7) required low-dose phenylephrine infusion (<0.5 μg/kg/minute) during the asleep phase. Patient 3 required continuation of phenylephrine during the resection; he also had total blood loss of 1350 mL. Nicardipine infusion was used in patients 4 and 7 to maintain the BP within the target range. Brain relaxation was sufficient in all cases.

All but 1 patient had complete resection, confirmed by immediate postoperative CT angiography/DSA. Patient 3 had residual AVM on follow-up CT angiography. He underwent a second successful awake resection, and postoperative angiography showed complete resection. Despite negative intraoperative motor mapping, patient 6 developed contralateral motor weakness, which was attributed to supplementary motor area syndrome and resolved completely within 1 week. All patients were discharged on postoperative day (POD) 4 except patient 6, who stayed until POD 6 because of his neurologic deficit. All patients underwent successful awake resection without significant intraoperative or postoperative complications. Improvement in neurologic status was seen in all patients without new deficits (Table 2). All patients were followed for a minimum of 6 months except for patient 7, who was doing well at 10 days. Patients did not report any negative experiences with the awake craniotomy and reported that they felt prepared to undergo awake procedures, if needed in the future. The pediatric patient did report some fatigue and anxiety intraoperatively, which is understandable given the long duration of the surgery.

Case Illustrations

Case 1 (Patient 2). A 50-year-old man with a history of depression and smoking presented with gradual onset of occipital headache, neck stiffness, photophobia, nausea, difficulty with reading comprehension, and aphasia. CT angiography and MRI showed a large left posterior temporal AVM (SMG 3) measuring 4.0 cm × 3.0 cm × 2.5 cm, located near the Wernicke area (Figure 1A). fMRI showed likely upward displacement of the inferior longitudinal or inferior fronto-occipital fasciculi by the AVM. Language fMRI showed bilateral activation of the Wernicke area, which was superior to the AVM in the left

hemisphere, and left hemisphere-dominant activation of the Broca area (Figure 1B and C).

The patient underwent cerebral angiography (Figure 1D) with successful embolization of feeding vessels with 60% nidal reduction, along with coiling of the intranidal aneurysm (3.5 mm) and a broad-based pedicle aneurysm (4.3 mm) under GA. Awake surgical resection of the AVM was performed the next day. The patient was sedated with the combination of propofol (starting dose 50 μg/kg/minute), remifentanil (starting dose 0.08 μg/kg/minute), and dexmedetomidine (starting dose 1.0 μg/kg/hour) infusions, gradually titrated to effect. Supplemental oxygen was provided by face mask. Vital landmarks, including the sagittal sinus, coronal sutures, motor strips, and speech areas, were defined. Baseline neurologic examination was documented and repeated throughout the surgery until gross total excision of the AVM was achieved. The AVM was noted to be large, complex, and heavily calcified, abutting the ventricle; total surgical time was 12 hours, with the awake phase 7.5 hours. The patient required light sedation toward the latter part of the excision because of shoulder pain, but he remained easily arousable and able to cooperate for the neurologic examination. Surgery was complicated by blood loss of 900 mL, requiring blood transfusion. Overall, the patient tolerated the procedure well. Postoperative CT head and cerebral angiography (Figure 1E) were negative for residual AVM or pedicle aneurysm, and at 1 year follow-up, he had improved reading ability, intact speech, and had experienced no seizures.

Case 2 (Patient 4). A 14-year-old girl with a history of large, left-sided parietotemporal hemorrhage presented with mild occasional headaches and slurred speech (Figure 2A–D). MRI showed a left temporal AVM measuring 2.5 cm × 0.6 cm located within the left middle cerebral artery territory. Language fMRI showed bilateral activation of the Wernicke area and left hemisphere activation of the Broca area, with activation in the left temporal lobe immediately adjacent to the region of encephalomalacia or hemorrhagic cavity. She was our youngest patient and had thorough counseling preoperatively. She was sedated with dexmedetomidine 1 μg/kg bolus, followed by a

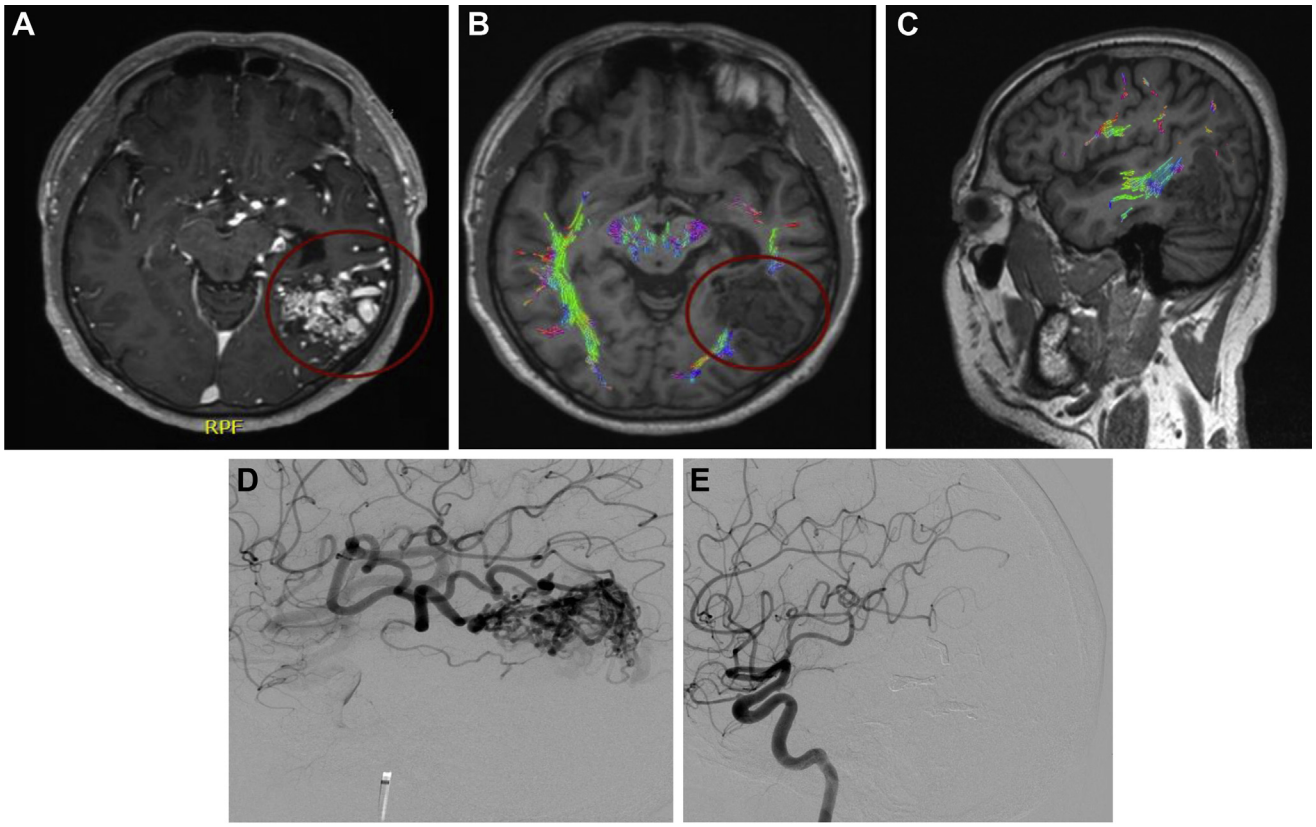


Figure 1. (**A**) Computed tomography angiogram head with contrast. Axial view of large arteriovenous malformation (*red circle*) located in left Wernicke area with some regional mass effect seen with mild dilatation of adjacent temporal horn of left lateral ventricle. (**B**) Functional magnetic resonance imaging of the brain, Language functional magnetic resonance imaging showing bilateral activation of Wernicke area, greater on the right, which is superior to the arteriovenous malformation (*red circle*) in left hemisphere. (**C**) White matter tracts close to the arteriovenous malformation. (**D**) Left posterior temporal arteriovenous malformation, with feeding vessels from the left posterior cerebral artery and distal left middle cerebral artery branches, and a large draining vein into the lateral left transverse sinus. (**E**) Postoperative digital subtraction angiography showing complete resection.

dexmedetomidine infusion (starting dose 0.7 μg/kg/hour) and propofol infusion (starting dose 200 μg/kg/minute), gradually titrated to effect, along with intermittent fentanyl boluses. She required a higher dose of sedatives, compared with our older patients. During the awake portion of the surgery, dexmedetomidine was continued at a low dose of 0.2 μg/kg/minute. Neurologic examination was repeated throughout the surgery and remained normal. However, the patient felt anxious and nauseous toward the latter part of the resection, requiring deeper sedation for the remainder of the surgery. Postoperative angiography confirmed complete resection; the patient recovered completely and had resumed cheerleading at school at 1 year follow-up.

Case 3 (Patient 6). A 22-year-old man with history of complex partial seizures and asthma presented with intermittent headaches. CT and MRI showed SMG 2 AVM, measuring approximately 2.5 × 2.1 cm, and closely associated with the precentral gyrus. fMRI showed foot and toe motor activation posterior and medial to the AVM, and finger activation lateral to AVM (Figure 3A and B). The day before resection, the patient underwent preoperative DSA (Figure 3C) and embolization under GA with 70% nidal reduction. He had risk factors for OSA and required bilateral nasal airway insertion to maintain airway patency during deep sedation, using our anesthetic protocol (Tables 2 and 3). Mapping showed motor cortex posterior to the AVM nidus. Arterial feeders were controlled after confirmation with temporary occlusion. The entire AVM nidus was resected, with preservation of draining vein, until the end. The patient remained neurologically stable throughout; however, during the last stage of AVM nidus mobilization, he experienced weakness of the left arm and leg, which progressively worsened. Complete

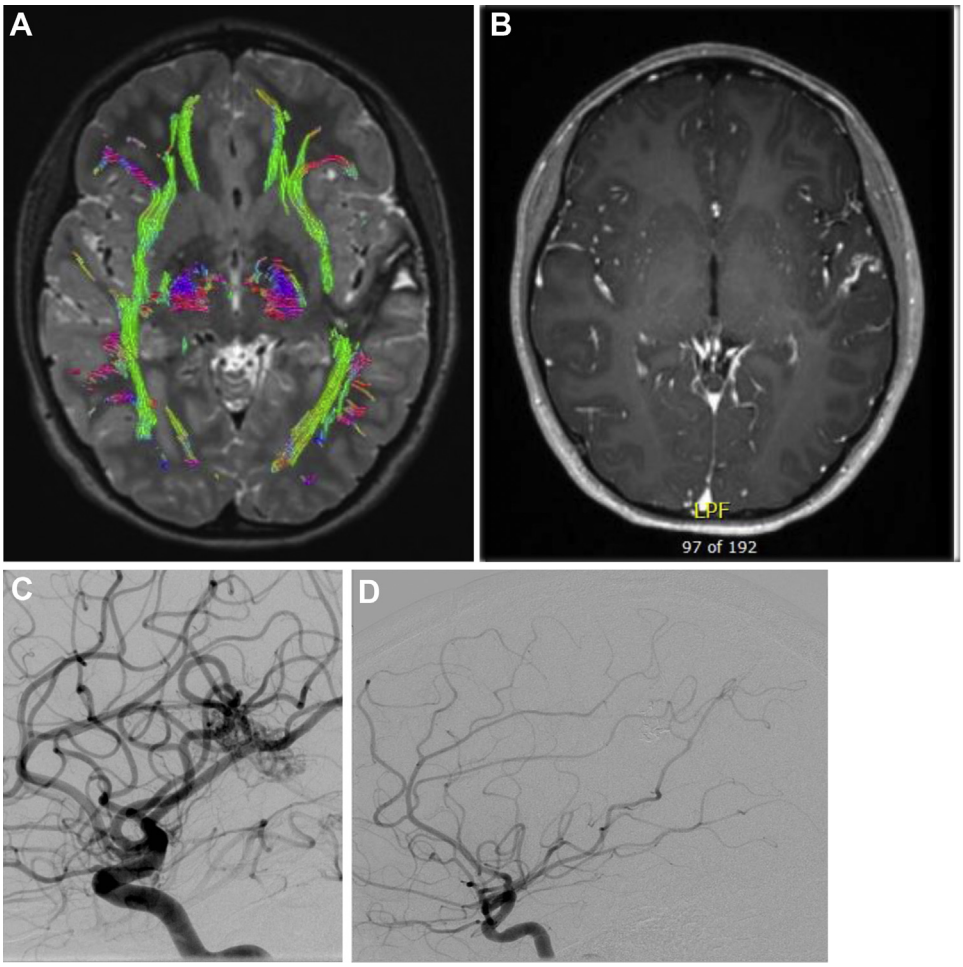


Figure 2. (**A, B**) Functional magnetic resonance imaging of the brain showing loss of fractional anisotropy in superior temporal encephalomalacia/hemorrhagic cavity with disruption in left inferior longitudinal fasciculus/inferior fronto-occipital fasciculus. (**C**) Preoperative cerebral angiogram showing arteriovenous malformation involving the Wernicke area, fed by middle cerebral artery branches. (**D**) Postoperative angiogram confirming complete resection.

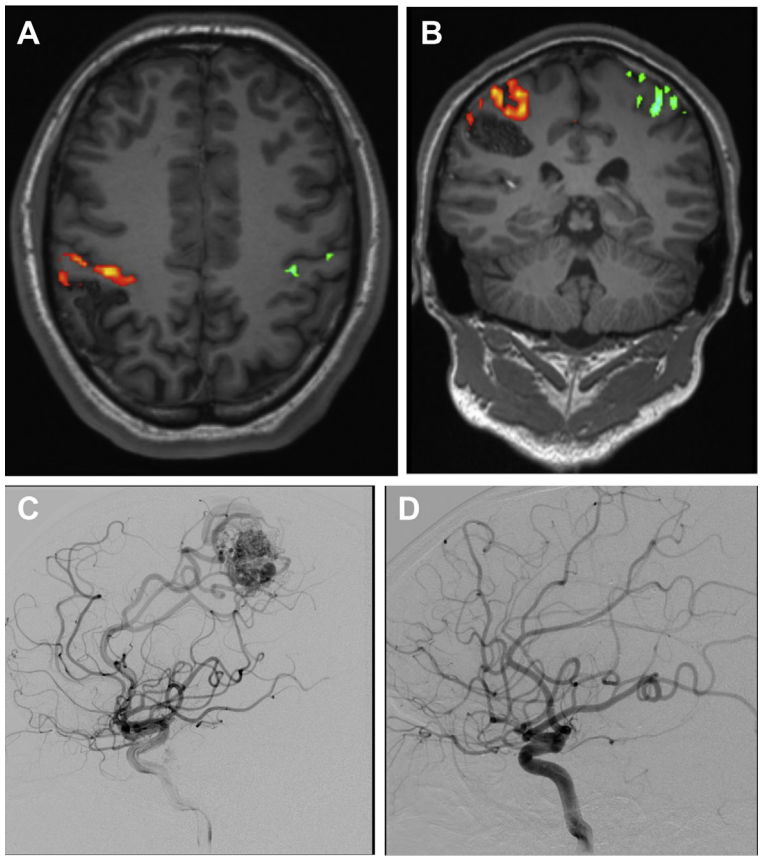


Figure 3. (**A, B**) Arteriovenous malformation, measuring approximately 2.5 × 2.1 cm, closely associated with the right precentral gyrus. Functional magnetic resonance imaging showing motor activation close to the arteriovenous malformation. (**C**) Digital subtraction angiography showing 2.5-cm right paramedian posterior frontal vascular nidus with supply from the pericallosal branch of the right anterior cerebral artery and early venous drainage via cortical vein into the superior sagittal sinus. (**D**) Postoperative digital subtraction angiography showing complete resection.

resection was performed because the AVM was near totally mobilized at that stage.

Postoperative cerebral angiography (Figure 3D) was negative for residual AVM and the patient slowly regained his strength over 1 week and was discharged home on POD 6. At 6 months follow-up, he had remained seizure free and had regained his strength.

Case 4 (Patient 7). A 50-year-old man with history of hypertension, smoking, drug and alcohol abuse, gastroesophageal reflux disease, and seizures presented with right-hand intermittent weakness/numbness, and intermittent speech difficulties. He had 2.5 × 2.3 cm left frontoparietal SMG 2 AVM with drainage via 2 large cortical veins. He also had 5.5 × 2.5 mm left internal carotid artery ophthalmic segment aneurysm and 3.0 × 2.0 mm anterior communicating artery aneurysm (Figure 4A and B). He was unable to undergo preoperative embolization because of the presence of aneurysms. His anesthetic management was challenging because of his history of alcohol and drug abuse and associated brain aneurysms. We followed our anesthesia protocol along with nicardipine infusion, for strict BP control. He successfully underwent awake resection of the AVM without any complications. Postoperative DSA showed complete resection (Figure 4C). He was discharged home on POD4.

DISCUSSION

Awake Craniotomy

Although sophisticated noninvasive functional imaging modalities can facilitate localization of the eloquent area, preoperative functional imaging serves only as a screening test, because there have been reports of inconsistent correlations between the functional imaging and DCS mapping.[23,24] Altered hemodynamics of

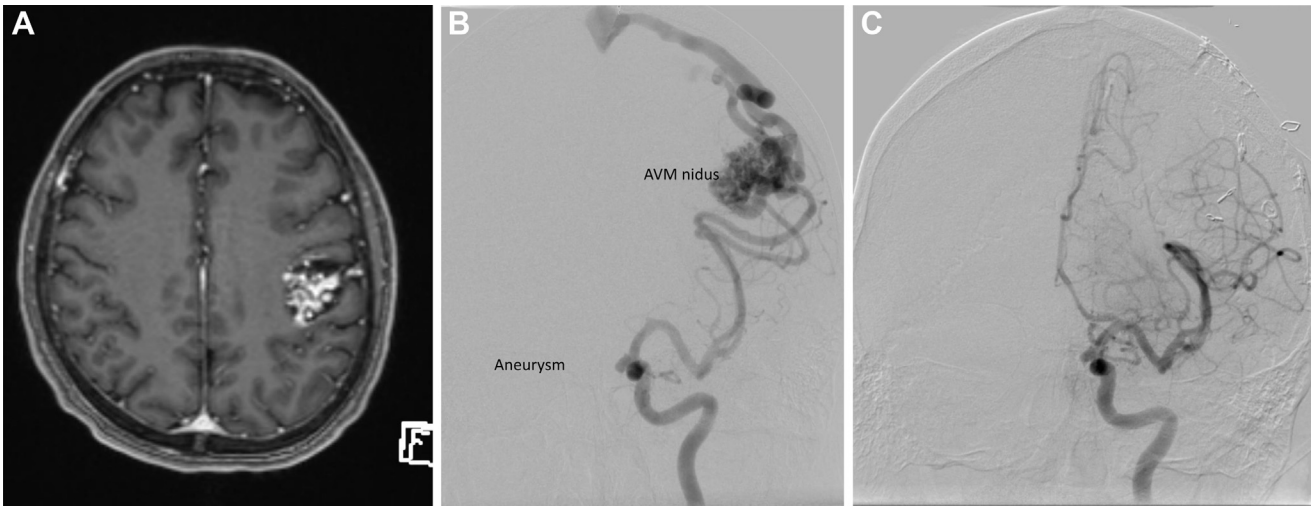


Figure 4. (**A**) Magnetic resonance imaging showing left frontoparietal arteriovenous malformation (AVM) with a 2.5 × 2.4 cm nidus which involves left postcentral gyrus and left parietal lobe. (**B**) Digital subtraction angiography showing 2.5 × 2.3 cm left frontoparietal Spetzler-Martin grade 2 AVM with drainage via 2 large cortical veins and fed from the branches of left middle cerebral artery, and 5.5 × 2.5 mm left internal carotid artery ophthalmic segment aneurysm. (**C**) Postoperative digital subtraction angiography showing complete resection of the AVM.

vascular lesions can interfere with functional imaging.[17,25] Long-standing lesions, including AVMs, have been shown to trigger significant shifts in motor or language function, from their anatomic locations to adjoining gyri or even to the contralateral hemisphere.[26,27] DCS and intraoperative neuromonitoring have been used successfully to guide surgical resection under GA.[17,28,29] However, the functional reorganization of the motor cortex near the lesion may limit the reliability of intraoperative cortical mapping alone for detecting the motor area.[30-32] This finding suggests that both brain mapping and awake neuromonitoring should be used during the resection. Awake patients allow for more precise brain mapping and superior clinical neurologic monitoring, which facilitates resection by defining the safe margins.

The superiority of awake craniotomy is well established for resection of intrinsic brain tumors in or adjacent to the eloquent areas. Meta-analyses have confirmed that awake surgery, as opposed to GA, facilitated more extensive tumor resection, along with a decrease in hospital stays, postoperative neurologic deficits, and surgical times.[15,33] In addition to the usual problems associated with awake craniotomy for tumor surgery, vascular lesions pose additional challenges. They are more prone to significant bleeding, which can be intimidating in an awake patient and may require emergent conversion to GA. Two in this series had significant bleeding (900 and 1350 mL); however, they were managed without conversion to GA. Some of the high-grade AVMs have diffuse margins with ambiguous dissection planes that might not be amenable to complete resection.[17] Awake surgery for AVMs does not allow for brain relaxation by hyperventilation or pharmacologic neuroprotection via burst suppression, which should be taken into consideration before proceeding.

Review of Literature

A literature review found 9 reported case series,[17-22,29,34,35] in which awake functional mapping was used for AVM/cavernoma resection. However, only 6 of these case series kept patients awake during the resection[17-22](**Table 4**). Burchiel et al.[34] first reported the technique of awake functional mapping, followed by resection under GA, in a series of 8 patients with AVMs of the sensorimotor and language cortex. Total resection was achieved without causing additional neurologic deficits in 7 of the 8 cases.

Cannestra et al.[18] presented a classification system for grading AVMs proximal to the language cortex into 3 levels of surgical risk. Group I had fMRI activation in ≥1 gyrus removed from the nidus and were considered minimal risk, this group underwent resection under GA. Group II had AVMs intimately associated with language function and were deemed high risk, opting for radiosurgery or observation. Group III (moderate risk) showed functional activation within 1 gyrus of the nidus without overlap and therefore underwent awake resection. Five moderate-risk patients were selected for awake resection with cortical mapping. In 2 of these patients, despite fMRI-suggested operability, cortical stimulation and optical imaging disclosed eloquent cortex encompassing the nidus, resulting in abortion of the resection, casting doubt on the true reliability of fMRI. These 2 patients had postoperative radiosurgery, with the resolution of the lesion in 1 patient. However, the second patient experienced a hemorrhagic event. The other 3 patients underwent complete resection, with 2 experiencing transient speech deficits that resolved within 3 months.

The technique of awake AVM excision was further explored by Gamble et al.,[19] who used awake mapping in conjunction with continued cortical and subcortical stimulation during

Table 4. Review of Past Case Series in which Awake Resection or Arteriovenous Malformation or Cavernoma Was Used Along with Cortical Simulation Mapping of Language or Motor Functions

	Awake Arteriovenous Malformation (Number of Patients)	Spetzler-Martin Grade	Cavernoma (Number of Patients)	Resulted in Subtotal Resection	Aborted Resection	Immediate Postoperative Deficits or Complications	Long-Term Deficits
Cannestra et al., 2004[18]	5	2 grade 2, 2 grade 3, 1 grade 4	0	0	2	2 (transient speech deficits)	1 (hemorrhagic event in 1 patient who had radiosurgery)
Gabarrós et al., 2011[17]	5	4 grade 4, 1 grade 4		1 had radiosurgery	0	3 (1 intracranial hemorrhage, 1 extradural hematoma, 1 hemorrhage)	1 new hemorrhage after radiosurgery
Gamble et al., 2015[19]	4	2 grade 2, 2 grade 2, 2 grade 3	0	0	0	1 (transient speech deficits)	0
Aoun et al., 2017[20]	2	2 grade 2	3	0	0	0	0
Matsuda et al., 2012[21]	0	Not available	9	0	0	2	2 patients with recurrent seizures that resolved later
Zamorano et al., 1998[22]	6	5 grade 1 or 2, 1 grade 3	0	0	0	0	0
Our series	6	5 grade 2, 1 grade 3	1	1	0	2 (facial weakness for 3 days in patient 3; left hemiplegia lasted for a week in patient 6)	0

resection in 4 patients with left-sided language zone AVMs. One patient developed a speech deficit during resection and subcortical stimulation, despite negative cortical mapping. This patient experienced mild temporary postoperative expressive dysphasia. Complete resection was achieved in all 4 patients, without any permanent neurologic morbidity. Thus, subcortical stimulation and awake language interrogation were critical in defining the safe margins and protecting the sensitive zone.

In contrast, Gabarrós et al.[17] used awake mapping and resection in 5 patients with AVMs near language areas, who were all grouped in the moderate-risk category (as defined by Cannestra et al.[18]). Their series also included 7 patients with AVMs near the motor cortex who underwent mapping and resection under GA. Awake mapping was deemed useful in all cases and influenced the surgical approach in 2 of the awake patients. Mapping limited resection in 1 patient with an SMG 4 AVM, who received postoperative radiosurgery but had a poor functional outcome secondary to hemorrhage.

Aoun et al.[20] described 2 patients with AVMs and 3 patients with cavernomas who underwent awake brain surface and white matter mapping before and during microsurgical resection. Matsuda et al.[21] reported 9 patients with cavernomas in the eloquent areas who underwent awake resection aided by cortical and subcortical mapping. These investigators reported 2 incidents of discovering language areas using subcortical mapping in the pericavernomatous gliosis, which prevented full removal of gliosis and spared eloquent brain tissue.

Although scientific evidence based on large randomized studies justifying surgical treatment of eloquent vascular lesions is lacking, results from these clinical series overwhelmingly support awake craniotomy as a safe treatment option.

Selection Criteria and Patient Preparation

Selecting ideal candidates for awake surgery must be based on thorough preoperative assessment. General contraindications for awake procedures include uncooperative patients, severe OSA, severe language disabilities, large and deep AVMs, and posterior fossa AVMs. The potential benefits of awake resection must be weighed against the possible intraoperative risks.

The patients must be prepared by establishing a trusting patient–physician relationship, with detailed assessments, reassurance, and counseling. Patients should be well informed regarding positioning, mapping tests, motor and/or language testing, and side effects, including dry mouth, nausea, and pain during surgical manipulation.

Intraoperative Management. Awake craniotomy for AVM is an uncommon procedure, which requires an expert team of

neuroanesthesiologists, neurosurgeons, operating room staff, and a motivated patient. Intraoperative emergence and the ability to interact constantly during mapping and neurologic evaluation play a vital role, highlighting the significance of anesthetic management and patient cooperation. Vigilance and thorough planning are paramount in anticipating and handling any untoward complications. The need to secure the airway emergently, possible nausea and vomiting, patient movement, emergence agitation, and delirium can challenge the anesthesia provider caring for a patient whose head is fixed in pins. Even mild hypoxemia or hypercarbia can impair brain relaxation and surgical exposure. Uncontrolled bleeding and sudden hemodynamic changes may require massive fluid or blood resuscitation, along with pharmacologic manipulation of the circulation, and emergency conversion to GA. Intraoperative seizure is another common complication, which is often treated with iced saline to the surface of the brain and rarely requires antiepileptic medications. Attention to patient positioning with proper padding of all pressure points is crucial, because staying in 1 position for a long period can be tiring and uncomfortable.

Anesthetic Technique. A variety of anesthetic techniques have been used for awake craniotomy,[15,36-38] ranging from the AAA technique, which involves GA with or without a definitive airway before and after the testing and resection phase; conscious sedation or monitored anesthetic care (MAC), in which patients are mildly sedated without the need for airway intervention; or fully awake patients, with only local or regional anesthesia of the scalp. The advantages of AAA include superior comfort for the patient and surgical team during the asleep phase, ensuring adequate ventilation, analgesia, and brain relaxation by hyperventilation and immobilization. However, risks include emergence delirium, slow awakening, increased somnolence, coughing, laryngospasm, and hypertension during extubation. In contrast, the MAC or fully awake technique limits the sedative burden, which precludes emergence delirium, the need for airway manipulation, or other complications of GA. However, it carries the risk of untoward patient movement, discomfort, and hypoxia. We preferred to use AAA technique without a definitive airway (endotracheal tube or laryngeal mask airway). We inserted nasopharyngeal airways as needed to maintain the airway patency, which allowed for deep sedation while avoiding patient movement, coughing, or emergence delirium. Despite significant differences in these 2 approaches, both AAA and MAC have been used successfully for intraoperative brain mapping without significant difference in the outcome, and the choice is often is based on the preferences of the anesthesiologist and surgical team.[36,38]

Drug Selection. The sedation and analgesic needs fluctuate throughout the different phases of surgery. Finding the optimal balance between dosage and type of sedative, analgesic, and anxiolytic medications, without cardiorespiratory compromise, is challenging. Various intravenous sedative medications (e.g., propofol, fentanyl, remifentanil, and dexmedetomidine) have been used for awake craniotomies.[15,36-40]

The propofol-based AAA protocol is highly efficient because of its rapid titratability and its antiepileptic and antiemetic effects.[36-38] However, it can cause dose-dependent respiratory depression, hypercapnia, and subsequent brain swelling. The safety and efficacy of dexmedetomidine sedation as a substitute to propofol during awake craniotomy have been evaluated in various clinical trials.[38,39] With its sedative, analgesic, and anesthesia-sparing effects, dexmedetomidine can be used as a sole agent or as an adjunct to other medications. Remifentanil carries favorable pharmacokinetics with its easy titratability and rapid dissipation.[38-40] Most of the earlier studies on awake craniotomies used either dexmedetomidine or propofol as their main sedative.[36-41] Our combination of low-dose infusions of propofol, dexmedetomidine, and remifentanil allowed us to achieve deep sedation during the asleep phase. Remifentanil was titrated to a goal respiratory rate of 8–12 breaths per minute. This regimen allowed us to keep the dosage of all 3 drugs at a lower level, thus reducing the risk of respiratory depression and need for airway interventions and facilitating rapid and smooth recovery after stopping the infusions. In some patients, low doses of remifentanil and/or dexmedetomidine were resumed during the resection to keep the patient comfortable but still able to cooperate for neurologic evaluation.

In addition to intravenous sedative infusions, we performed a scalp block and infiltration of local anesthetic along the incision and pin sites. Long-acting local anesthetics (bupivacaine and ropivacaine) can provide reliable analgesia lasting for several hours and limit the need for opioids. Low-dose remifentanil infusions can be used to supplement sedative medications during the asleep and awake phases, because some patients can still have unpleasant sensations during perivascular traction, despite adequate scalp block.

Study Limitations

Limitations of our study include the retrospective design and small sample size. Most of our patients belonged to SMG <3, which is similar to previous case series (except in the study by Gabarrós et al., in which in all 5 patients showed SMG ≥ 3). Hence, usefulness of this technique in patients with high-grade AVM (≥ 3) needs to be evaluated. Complete surgical resection may not be feasible in some complex AVMs because of closeness to the eloquent areas. As described by Gabarrós et al.[17] and Cannestra et al.,[18] intraoperative mapping resulted in subtotal resection in 1 patient and abortion of the procedure in 2 others. These complex AVMs and other high-grade AVMs are often managed by radiosurgery.[17,18] Two of these patients experienced intracranial hemorrhage, because radiosurgery often does not achieve immediate obliteration, placing these patients at risk for bleeding from the residual nidus.[17,18] However, our study still significantly contributes to the literature, which confirms the safety and efficacy of awake craniotomies and microsurgical resection of intracranial vascular lesions in carefully selected patients.

CONCLUSIONS

Awake craniotomy is a feasible option in carefully selected patients with intracranial AVMs and cavernomas. Our anesthesia protocol of combining low-dose infusions of propofol, remifentanil, and dexmedetomidine was effective in achieving deep sedation for the asleep phase, allowing for smooth

emergence and optimal patient cooperation during the awake phase, with no significant perioperative complications or need to convert to GA.

ACKNOWLEDGMENTS

We would like to thank Dr. Carin Mascetti, M.D. for proofreading the article.

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Conflict of interest statement: The authors declare that the article content was composed in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Received 26 May 2020; accepted 9 July 2020

Citation: World Neurosurg. (2020) 143:e136-e148. https://doi.org/10.1016/j.wneu.2020.07.050

Journal homepage: ***www.journals.elsevier.com/world-neurosurgery***

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